

Probability before Kolmogorov

Section 5 · Live writing edition

Soma S Dhavala · IIT Jammu

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What does a probability mean?

In a hypothetical agent benchmark, success was recorded on 90 of 100 tasks. What does the statement “success probability is 0.9” assert, and what does it assume?

The calculus and its interpretation

Separate two questions: which probability calculations are valid, and what do the numbers represent?

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Laplace's classical rule

A fair die has six equally likely faces. What probability does the classical rule assign to an even face?

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The assumption behind counting

What justifies assigning the same probability to every face?

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A circular definition

Can “equally possible” define probability without already using a probability idea?

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One experiment, two partitions

Toss a fair coin twice. What is the probability of at least one head? First list the ordered outcomes.

The same event with three cases

A three-case account lists: a head on the first toss; a tail then a head; two tails. Are these cases equally likely?

Partition dependence

What failed in the three-case calculation?

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When the counting rule is valid

State a condition under which $P(A) = |A|/|\Omega|$ is justified for finite Ω .

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A measured fraction is not a probability

In n observed trials, the event A occurred $N_n(A)$ times. What does $N_n(A)/n$ describe?

The frequency interpretation

How does the frequency account define the probability of A ?

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What is a collective?

Why did von Mises impose a condition on selected subsequences?

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Limits of the frequency account

Can a finite run establish that the limiting relative frequency exists? What about a one-time event?

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A theorem or a definition?

Under the frequency interpretation, is convergence a theorem? Under the axioms, what is its status?

Probability as degree of belief

How might a probability describe uncertainty about an event that will not be repeated?

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Coherence constrains beliefs

What condition prevents a system of betting probabilities from guaranteeing a loss?

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Finite and countable additivity

Does coherence alone justify countable additivity?

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Probability as a relation to evidence

Can probability be understood as a relation between evidence and a proposition, rather than a personal belief?

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The principle of indifference returns

How does the logical account face the difficulty of assigning probabilities without enough evidence?

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Fiducial probability

What was Fisher's fiducial argument intended to obtain?

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What fiducial probability is—and is not

Is fiducial probability an alternative interpretation of the probability calculus?

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Bertrand: a question without a sampling rule

If someone says “choose a chord at random,” is the probability model specified?

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What Bertrand does not explain

Does specifying a sampling rule provide a framework for proving results about infinite sequences or processes?

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Measure theory preceded the axioms

Was measure theory invented to resolve Bertrand's paradox?

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Infinite sequences need measurable events

To say that relative frequencies converge with probability one, what set must be an event?

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Processes on function spaces

What kind of probability space can represent a random continuous path?

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Conditioning on a probability-zero event

Why does $P(A \mid B) = P(A \cap B)/P(B)$ not define conditioning when $P(B) = 0$?

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Axiomatization and mechanics

Was probability axiomatization also part of a broader mathematical programme?

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What the axioms settle

Do the axioms determine which probability interpretation is correct?

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Reading guide

Which arguments should be revisited in the main notes, and which belong to the Bertrand supplement?